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Trust my IDS: An explainable AI integrated deep learning-based transparent threat detection system for industrial networks

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Abstract

Industrial networks are vulnerable to various cyber threats that can compromise their Confidentiality, Integrity, and Availability (CIA). To counter the increasing frequency of such threats, we designed and developed an Explainable Artificial Intelligence (XAI) integrated Deep Learning (DL)-based threat detection system (XDLTDS). We first employ a Long-Short Term Memory-AutoEncoder (LSTM-AE) to encode IIoT data and mitigate inference attacks. Then, we introduce an Attention-based Gated Recurrent Unit (AGRU) with softmax for multiclass threat classification in IIoT networks. To address the black-box nature of DL-based IDS, we use the Shapley Additive Explanations (SHAP) mechanism to provide transparency and trust for the system's decisions. This interpretation helps SOC analysts understand why specific events are flagged as malicious by the XDLTDS framework. Our approach reduces the risk of sensitive data and reputation loss. We also present a Software-Defined Networking (SDN)-based deployment architecture for the XDLTDS framework. Extensive experiments with the N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets confirm the effectiveness of XDLTDS against existing frameworks in addressing modern cybersecurity challenges and protecting industrial networks.

Introduction

The Internet of Things (IoT) is a transformative communication approach representing a mutually connected pattern of physical objects, devices, sensors, and protocols to formulate a smart dataexchanging framework (Rathee et al., 2022). IoT systematically enables an automated networking architecture where the sensors interlink the participant nodes, and the communication protocol regulates the data transmission, consequently mitigating the need for direct human interaction (Ashraf et al., 2022). The concept is widely acknowledged, resulting in extensive growth of IoT applications in every sector, including smart homes, smart cities, smart healthcare, smart agriculture, smart transportation, smart monitoring systems, etc. The industrial domain is more likely privileged by IoT and reveals a vast variety of IoT applications in the industrial scenario, referred to as the Industrial Internet of Things (IIoT). Over the past decade, IIoT has significantly revolutionized the typical paradigm for industrial systems by introducing a fusion of innovative technologies into conventional industrial processes (Adil et al., 2022). The phenomenon is interpreted by the literature term Industry 4.0, yielding numerous benefits, e.g., smart manufacturing, communication reliability, resource optimization, predictive maintenance, increased efficiency, enhanced safety measures, and many more (Chiu et al., 2022). The increasing demands of IIoT and the flourishing circle of its applications experience the involvement of heterogeneous communication nodes interlinked by cyber links over the internet. Such multidimensional heterogeneous connectivity becomes a potential ground for bad actors to perform adversarial activities to compromise the integrity of the system. The examples must include zero-day exploits, Cross-Site Scripting (XSS), Structured Query Language (SQL), and Man-in-the-Middle (MITM) attack categories (Li et al., 2022). Counting these categories, the IIoT systems may also become vulnerable to a broader range of other security challenges, such as Denial of Service (DoS) attacks, crypto-jacking attacks, replay attacks, side-channel attacks, device spoofing attacks, cloud-based attacks (Chen et al., 2020), etc. Along with the security breaches, the threats also prohibit IIoT systems from meeting operational sustainability, efficient scalability, and regulatory compliance. Such circumstances flag an alarming situation and require significant security countermeasures to be adopted.

Artificial Intelligence (AI) provides a productive insight into developing security solutions to address the mentioned challenges. Deep Learning (DL) is an AI-driven pathway increasingly implied to design more sophisticated security frameworks for IIoT. In recent years, DL-empowered Intrusion Detection Systems (IDS) have attained considerable attention since their complex computational cycles in decision-making, leading to substantial filtering between normal and suspicious scenarios (Lo et al., 2022). DL-based IDS is initially trained on a dataset containing the impressions of threats and then employed to identify malicious activities from regular ones. Undoubtedly, DL-driven IDS are a remarkable choice for the timely detection of malicious entities in IIoT communications. However, there is still ample room for improvement to boost the reliability and trustworthiness of these security methods (Javeed et al., 2023b). Traditional DL approaches adopt a black-box strategy where all the internal processing of algorithms is hidden, making it challenging for humans to understand their decision-making process. While working in sensitive industrial environments, it becomes crucial for security analysts to consider the logic behind declaring a specific activity as malicious (Ahmed et al., 2022). This context raises unaddressed questions regarding the steps involved in the decision-making processes of DL approaches, prompting the need for some transparent solutions. Explainable Artificial Intelligence (XAI) is an emerging approach that refers to an absolute package of methods and techniques to interpret the decision-making process of DL algorithms (Rawal et al., 2021). XAI explores the black-box concept and explains the steps contributing to the decision-making process, providing humans with a clear understanding of the logic employed to flag a particular event as malicious. XAI facilitates conventional DL algorithms with transparency, interpretability, trustworthiness, and reliability. XAI also makes the DL approaches accountable for making certain decisions, reducing the probabilities of inefficiencies and biased decision-making (Kök et al., 2023). While addressing high-stakes industrial environments, it becomes comparatively more indispensable to interpret the decisions taken by automated systems concerning regulatory compliance. Integrating XAI in DL-driven IDS is an impressive idea for comprehending decision-making and transparently considering the intrusion detection process.

IIoT is revealing manifestation in every sphere of our lives and its application circle is remarkably expanding to assist disparate aspects of our lifestyle. While providing several benefits, it is also vulnerable to a myriad of cyber threats that can lead to severe consequences, i.e., equipment damage, production disruptions, data breaches, and data loss. A typical IoT-based industrial network may be a susceptible victim to a variety of cyber-attacks, i.e., botnet attacks, spoofing attacks, Denial of Service (DoS), eavesdropping, and Distributed DoS (DDoS) attacks (Zainudin et al., 2022). Consequently, the IIoT demands an efficient Threat Detection System (TDS) to maintain its functionality and integrity. Designing a TDS for the IIoT networks is of paramount importance to ensure the privacy, stability, and security of its critical processes. Numerous researchers have suggested ML and DL-based IDSs in recent years to protect the IIoT network from such virulent threats and attacks (Javeed et al., 2023a). However, such models are frequently prone to overfitting and are considered black boxes due to their lack of interpretability. Thus, the IIoT is still in dire need of an efficient, transparent, and resilient framework to secure its network against sophisticated threats. Therefore, we will consider the following Research Questions (R) in this research work:

- **(R1): How to develop an efficient and resilient IDS to detect the sophisticated threat in such a network with high accuracy and low false alarm rate?** The existing signature-based IDSs mostly rely on familiar attack signatures and have notable false negative and false positive rates, which limits their capability of detecting unknown threats. Although these signature-based IDSs are known to be effective, they may be outsmarted by the attackers using various ways. Consequently, it makes them impractical for deployment.

- **(R2): How to provide a solution for IIoT's heterogeneous nature and develop a scalable and centralized IDS that does not overburden the resource-constrained IIoT devices?** The IIoT networks have distinctive properties and require different threat detection mechanisms than the traditional networks. It consists of a considerable variety of resource-constrained and heterogeneous devices following different protocols, where each device requires diverse security measures. Further, IIoT networks frequently scale quickly with new devices constantly being added.

- **(R3): How to develop a transparent Threat detection System and interpret its decision for enhanced Trust?** The lack of transparency of the DL-based IDSs is one of the reasons for their vigilant adoption in critical systems. The security analysts consider such model black boxes as they lack the interpretation of their decisions. Understanding the rationale behind a model decision-making process promotes trust, and confidence, enabling the Security Operation Center (SOC) analysts to confirm that it is efficiently and securely doing its job.

The following are the contributions of this research work.

- **To address R1,** we design a DL-based TDS to protect IIoT networks against evolving threats. The proposed framework is developed using Long-Short Term Memory-Auto Encoder (LSTM-AE) to encode the IIoT network data in order to prevent inference attacks. After that, we employ an Attention-based Gated Recurrent Unit (AGRU) with softmax for multiclass classification of attacks in such a network.
- **To address R2,** we proposed an SDN-based deployment architecture for the proposed TDS. We placed the proposed TDS into the CP of the SDN for the following reasons. Firstly, it empowers an adaptive, dynamic, and scalable security mechanism, which means we do not need to major architectural changes with the expansion of the network. Secondly, it aids Security Operation Center (SOC) analysts in identifying threats efficiently with low false alarm rates by minimizing unnecessary alerts, which minimizes the potential damage to the network. Thirdly, it optimizes resource usage and does not overburden the resource-constrained devices, thus ensuring the overall reliability and security of the IIoT networks.
- **To address R3,** we employ the SHAP mechanism of the XAI to interpret the feature of the datasets that primarily contributed to the predictions of the proposed IDS. This explainability of the decision will help the SOC analysts to better understand and analyze the decision. In addition, it will promote trust and reliability of the proposed framework. Consequently, our proposed framework will allow industries to defend against sophisticated threats to reduce the risk of sensitive data loss and reputation loss.

The rest of the paper is organized as follows; Section 2 discusses the relevant literature. Section 3 presents the proposed mechanism. Section 4 describes the experimental setup with evaluation metrics. Section 5 discusses the results and performance of the proposed XDLTDS framework. Section 6 discusses the XAI-based Interpretation. In Section 7, we provide a discussion. Finally, Section 8 concludes the paper.

Section snippets

Literature overview

Industrial network security has grabbed the attention of researchers for the past few years, and they proposed various threat detection mechanisms as shown in Table 1. For example, researchers in Ding et al. (2023) proposed a DL-based model named DeepAK-IoT for cyber threat detection. Their proposed model comprises three blocks, i.e., the RSR block comprising CNN, the TRB block comprising GRU and LSTM, and the DB block comprising softmax. The authors utilized TON-IoT, Edge-IIoT, and UNSW-NB15 …

Proposed mechanism

In this section, we provide the complete details of our proposed approach. First, we discuss the XAI-integrated DL-based threat detection module. Second, we present details about the Deployment Architecture.

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Implementation setup

The experiments were conducted using the Keras framework on the TensorFlow platform with Python version 3.8. We utilized the Spyder IDE for experimentation. For comprehensive details on the architecture and hyperparameters of the proposed model, please refer to Table 2. …

Datasets description

To evaluate our proposed framework, we employed three publicly available datasets, namely N-BaIoT (Meidan et al., 2018), Edge-IIoTset (Ferrag et al., 2022), and CIC-IDS2017 (Sharafaldin et al., 2018) datasets. We chose these …

Results & Analysis

In this section, we discuss the results and evaluate the performance of the proposed XDLTDS framework. First, we present and analyze its performance using standard evaluation metrics, followed by an assessment of the real-time detection feasibility of the proposed IDS. Additionally, we compare its performance with state-of-the-art mechanisms from the existing literature. …

XAI-based interpretation

This section presents the interpretability of the proposed XDLTDS mechanism by explaining the learned features of the data that contribute the most to the model's decision. We employed several plots provided by XAI to interpret the features' contributions to the predictions of the proposed XDLTDS framework, enhancing trust and transparency.

The Summary (SM) plot combines the significance of the features with their impacts. Each point on the SM plot represents an SV for an instance and a feature. …

Discussion

Detecting sophisticated threats in IIoT networks is a challenging task due to its heterogeneous and resource-constrained nature, where each device follows a different protocol and requires a different security mechanism. In the last few years, a variety of mechanisms has been proposed by the researchers. However, they encounter significant challenges while designing an efficient IDS for such a network due to the heterogeneous and resource-constrained nature of IIoT devices. These challenges …

Conclusion

In this paper, we develop the “XDLTDS” threat detection system to address the growing cybersecurity challenges in industrial networks. Our system leverages the combined strengths of XAI and DL techniques. The proposed system employs LSTM-AE to encode IIoT data, mitigating the risk of inference attacks, and uses the AGRU model with softmax for multiclass classification of cyber threats. To enhance transparency, we integrate the SHAP mechanism, which assists cybersecurity analysts in …

CRediT authorship contribution statement

Shifa Shoukat: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Tianhan Gao:** Writing – review & editing, Supervision, Formal analysis. **Danish Javeed:** Writing – original draft, Validation, Formal analysis, Conceptualization. **Muhammad Shahid Saeed:** Writing – review & editing, Writing – original draft. **Muhammad Adil:** Writing – review & editing, Validation, Formal analysis. …

Declaration of competing interest

The authors declare that they have no significant competing financial, professional, or personal interests that might have influenced the performance or presentation of the work described in this manuscript. …

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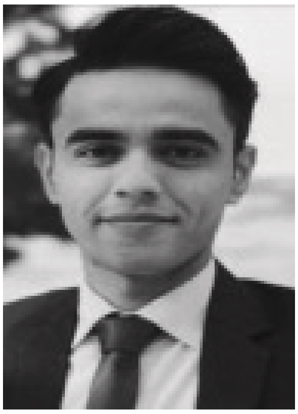

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